

# Statistics Crib Sheet

## Introduction

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\begin{aligned} s^2 &= \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{X})^2 \\ &= \frac{1}{n-1} \sum_{i=1}^n (x_i^2 - n\bar{X}^2) \\ s &= \sqrt{s^2} \\ &= \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i^2 - n\bar{X}^2)} \\ &= \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i^2 - n\bar{X}^2)} \end{aligned}$$

$$\begin{aligned} \bar{Y} &= a + b\bar{X} \\ S_Y^2 &= b^2 s_X^2 \\ S_Y &= |b|s_x \end{aligned}$$

$$pth \text{ percentile} = \frac{p}{100} (n+1)$$

- A histogram is perfectly symmetric if its right half is a mirror image of its left half.
- A histogram with a long right-hand tail is called skewed to the right or positively skewed.
- A histogram with a long left-hand tail is called skewed to the left or negatively skewed.
- A histogram is unimodal if it has only one peak, or mode. A histogram is bimodal if it has two peaks, or mode. A bimodal histogram, in some cases, indicates that the sample can be divided into two subsamples that differ from each other. If there are more than two peaks in a histogram, then it is said to be multimodal.
- Steps in construction of a boxplot

1. Computer the median and first and third quartiles of the sample. Indicate these with horizontal lines.
2. Find largest sample value that no more than  $1.5IQR$  above the third and quartile, and smallest value less than the first quartile.
3. Plot points. (first quartile)  $1.5IQR < x < 1.5IQR$  (third quartile)

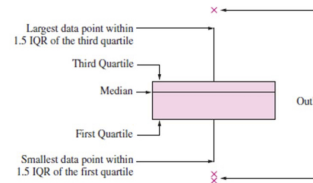


FIGURE 1.13 Anatomy of a boxplot.

## Probability

1. Let  $S$  be a sample space. Then  $P(S) = 1$
2. For any event,  $0 \leq P(A) \leq 1$
3. If  $A$  and  $B$  are mutually exclusive events, then  $P(A \cup B) = P(A) + P(B)$ .

## Complement Rule

**Complement Rule**  $P(A^C) = 1 - P(A)$

**Addition Rule**  $P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- The probability with equally likely outcomes has a probability  $P(A) = \frac{K}{N}$
- The number of permutation of  $k$  objects chosen from a group of  $n$  objects is

$$P_{n,k} = \frac{n!}{(n-k)!}$$

- The number of combinations of  $k$  objects chosen from a group of  $n$  objects

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

- The number of ways of dividing a group of  $n$  objects into groups of  $k_1, k_2, \dots, k_r$

$$\frac{n!}{k_1!k_2! \dots k_r!}$$

- A probability that is based on a part of a sample space called a conditional probability.

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- If  $A$  and  $B$  are two events with  $P(B) \neq 0$ , then

$$\begin{aligned} P(A \cup B) &= P(B)P(A|B) \quad \forall B \in \mathbb{R}, B \neq 0 \\ &= P(A)P(B|A) \quad \forall A \in \mathbb{R}, A \neq 0 \end{aligned}$$

- Let  $A$  and  $B$  be events with  $P(A) \neq 0$ ,  $P(A^C) \neq 0$ , and  $P(B) \neq 0$ .

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^C)P(A^C)}$$

Generally,

$$P(A_k|B) = \frac{P(B|A_k)P(A_k)}{\sum_{i=1}^n P(B|A_i)P(A_i)}$$

- The cumulative distribution function of r.v  $X$  is defined as:

$$F(X) = P(X \leq x) = \sum P(X \leq t) = \sum P(t)$$

- Let  $X$  be a discrete random variable with probability mass function  $p(x) = P(X \leq x)$  / The **Mean** is given by

$$\begin{aligned} \mu_x &= \sum xP(X = x) \\ &= \int_{-\infty}^{\infty} x f(x) dx \end{aligned}$$

- The **variance** of  $X$  is given by

$$\begin{aligned} \sigma_X^2 &= \sum (x - \mu_x)^2 P(X = x) = \sum x^2 P(X = x) - \mu_x^2 \\ &= \int_{-\infty}^{\infty} (x - \mu_x)^2 f(x) dx = \int_{-\infty}^{\infty} x^2 f(x) dx - \mu_x^2 \end{aligned}$$

- The median of  $X$  is the point  $x_m$  that solves the equation

$$F(x_m) = P(X \leq x_m) = \int_{-\infty}^{x_m} f(x) dx = 0.5$$

- The  $p$ th percentile is

$$F(x_m) = P(X \leq x_m) = \int_{-\infty}^{x_m} f(x) dx = \frac{p}{100}$$

- If a random variable  $X$  is multiplied by a constant  $a$  and then added to another constant  $b$ , then we have a new random variable  $Y$ , where

$$\begin{aligned} Y &= a * X + b \\ E(Y) &= \mu_y = a * E(X) + b \\ Var(Y) &= \sigma_y^2 = a^2 * var(X) \\ \sigma_Y &= |a|\sigma_x \end{aligned}$$

- If  $X_1, \dots, X_n$  is a simple random sample from a population with mean  $\mu$  and variance  $\sigma^2$ , then the sample mean  $\bar{X}$  is a random variable with

$$\begin{aligned} \mu_{\bar{X}} &= \mu \\ \sigma_{\bar{X}}^2 &= \frac{\sigma^2}{n} \end{aligned}$$

The standard deviation of  $\bar{X}$  is

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

**Chebyshev's Inequality** Let  $X$  be a random variable with mean  $\mu_x$  and standard deviation  $\sigma_x$ . Then

$$P(|X - \mu_x| \geq k\sigma_x) \leq \frac{1}{k^2}$$